

RAK13600 WisBlock NFC Reader Module Datasheet

Overview

Description

RAK13600 is a **WisBlock Wireless** NFC reader module based on the PN532 chip. It provides a compact and low-power solution for NFC-related IoT applications. The RAK13600 has an integrated transceiver module used for NFC applications which operates at 13.56 MHz and is also based on the standard 80C51 microcontroller core. It supports 6 different operating modes:

- ISO/IEC 14443A/MIFARE Reader/Writer
- FeliCa Reader/Writer
- ISO/IEC 14443B Reader/Writer
- ISO/IEC 14443A/MIFARE Card MIFARE Classic 1K or MIFARE Classic 4K card emulation mode
- FeliCa Card emulation
- ISO/IEC 18092, ECMA 340 Peer-to-Peer

Features

- **Module specifications**
 - Based on NXP PN532
 - External Antenna
 - Low power
- **Size**
 - 35 × 25 mm

Specifications

Overview

Mounting

The RAK13600 NFC reader module can be mounted to the IO slot of the WisBlock Base board. To illustrate, **Figure 1** shows how RAK13600 can be mounted on RAK5005-O WisBlock Base.

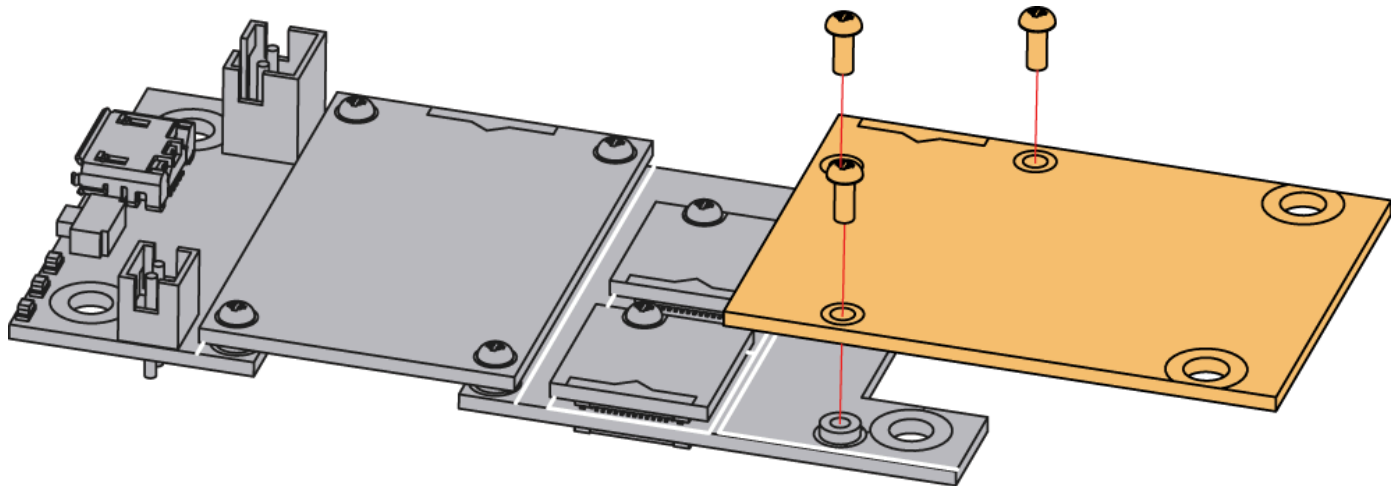


Figure 1: RAK13600 Mounting to WisBlock Base

Hardware

The hardware specification is categorized into four parts. It discusses the pinouts of the module and its corresponding functions. It also covers the electrical and mechanical parameters that include the tabular data of its characteristics. This section also shows the schematic diagram of RAK13600.

Chipset

Vendor	Part number
NXP	PN532

Pin Definition

The RAK13600 WisBlock NFC Reader has the standard 40-pin WisConnector. It is compatible with the IO slot which allows the RAK13600 module to be mounted on a WisBlock baseboard, such as RAK5005-O. The pin order of the connector and the Pinout Definition is shown in **Figure 2**.

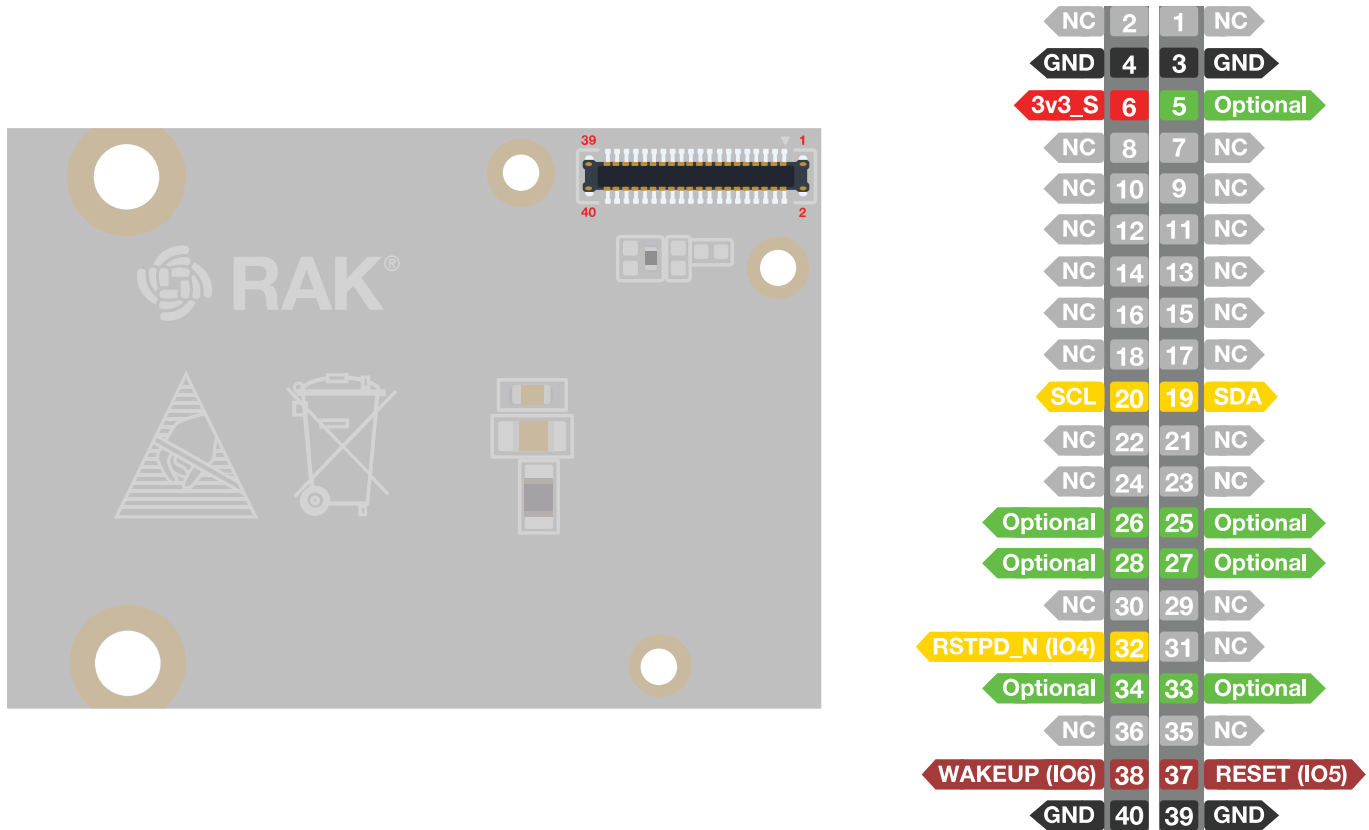


Figure 2: RAK13600 IO Slot Connector

NOTE

- The default interface is **I2C**.
- Reset functionality is available on **RESET (IO5)** and **RSTPD_N (IO4)** pins.
- PN532 wake up is available on **WAKEUP (IO6)**.
- **3V3_S** and **GND** supply power to the board.

Electrical Characteristics

Parameter	Minimum	Typical	Maximum	Unit
3V3_S	2.7	3.3	3.6	V
I _{dvdd} (Digital Supply Current)		25		mA
I _{avdd} (Analog Supply Current)		6		mA
I _{tvdd} (Transmitter Supply Current)		60	150	mA
Ambient Temperature	-30		80	°C

Mechanical Characteristics

Board Dimensions

Refer to **Figure 3** below for the mechanical dimensions of the RAK13600 module.

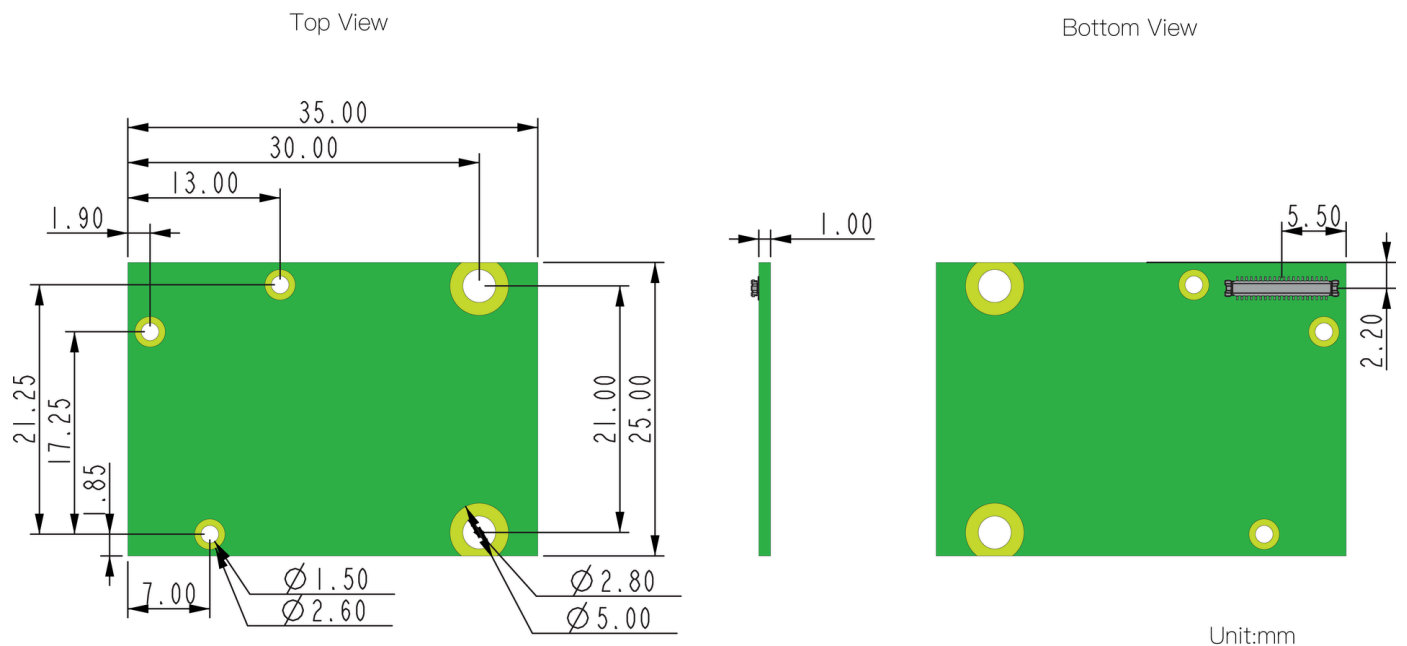


Figure 3: RAK13600 Mechanical Dimensions

WisConnector PCB Layout

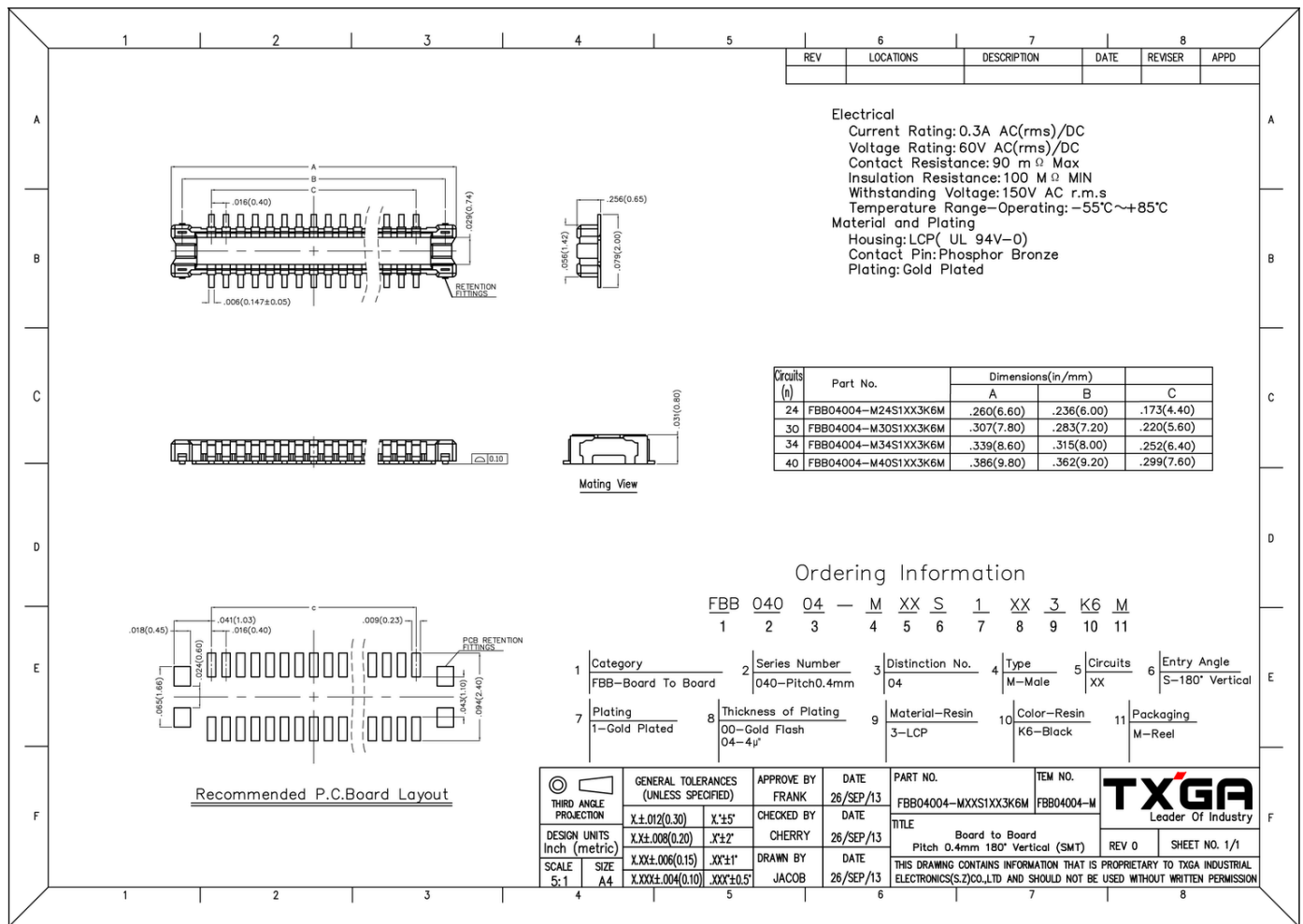
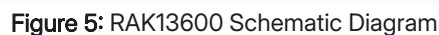


Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram



CE FCC

